

Safety Decoding in LLMs

Important papers

Language Model Unalignment

RESTA

Safety decoding in LLMs *[40 marks]*

[RESTA paper implementation]

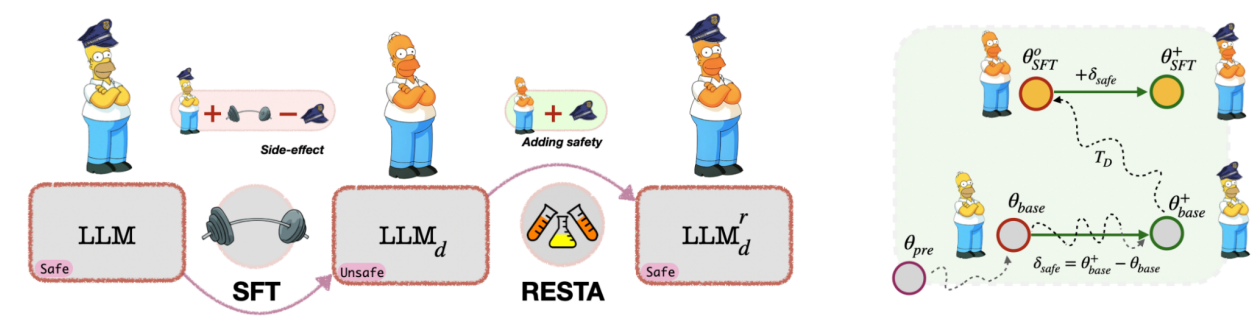


Figure: (Picked from RESTA paper) Representation of a brief idea of RESTA based safety decoding in LLMs

Objective_1: Comparing *unsafety scores* upon different strategies:

- Base Model $\rightarrow \theta_{\text{base}(+)}$
- SFT $\rightarrow \theta_{\text{sft}(0)}$
- SFT + DARE
- SFT + RESTA $\rightarrow \theta_{\text{sft}(0)} + \delta_{\text{safe}}$
- SFT + RESTA + DARE $\rightarrow \theta_{\text{sft}(0)} + \delta_{\text{safe}} + \text{DARE}$

Unsafety Evaluation criteria: This will be evaluated on the unsafety score = fraction of harmful responses. Lower is better.

Objective_2: Comparing the model performance on provided SFT dataset (see Step: 1 for SFT dataset) across different strategies:

- Base Model $\rightarrow \theta_{\text{base}(+)}$
- SFT $\rightarrow \theta_{\text{sft}(o)}$
- SFT + DARE
- SFT + RESTA $\rightarrow \theta_{\text{sft}(o)} + \delta_{\text{safe}}$
- SFT + RESTA + DARE $\rightarrow \theta_{\text{sft}(o)} + \delta_{\text{safe}} + \text{DARE}$

Performance Evaluation criteria: Use ROUGE-L, METEOR and BLEU metrics for this evaluation.

Note: Objective_1 is important for safety measures and **Objective_2** is necessary to assess the performance of your model in different setups.

(Step: 1) Supervised fine-tuning

Dataset: [sahil2801/CodeAlpaca-20k](#)

Goal: Obtaining $\theta_{\text{sft(o)}}$

Models from this step: SFT and SFT + DARE

You will have to fine-tune across two different strategies and will need to compare as well in your final results:

1. Parameter efficient fine-tuning (**PEFT**)
2. Full fine-tuning (**Full-FT**)

Based on the above dataset (**in Step: 1**), use the **first 40%** as your training data and the **last 10%** as your testing samples. Indirectly, you will be dealing with only **50%** data. Strictly follow this distribution for train-test.

You can design the prompt on your own. The employed prompt for fine-tuning and all hyper-parameters must be clearly mentioned in the **Report**. Note that your prompt should be based on *instruction* format for the best results. **Eg:**

Below is an instruction that describes a coding task, paired with an input that provides further context. Write a response that appropriately completes the request.

Instruction:

{instruction}

Input:

{input}

Output:

{output}

As the next step, apply **DARE** (see paper). DARE applies drop and re-scale to the delta parameters of the SFT model. This is done because an extensive set of SFT delta parameters are redundant and can be zeroed out; without impacting the model much. Your drop and re-scale must be based on Bernoulli's distribution. Use [mergekit](#) to finally get your SFT + DARE models.

(Step: 2) Unalignment and obtaining safety vector

Dataset: [unalignment/toxic-dpo-v0.2](#)

Goal: Obtaining δ_{safe}

Formula to get safety vector: $\delta_{\text{safe}} = \theta_{\text{base}(+)} - \theta_{\text{base}(\sim)}$

Here, you will be implementing an approach which slightly differs from what is done in the paper. Instead of fine-tuning on samples from test datasets (see Step: 4), you will be selecting the samples based on the dataset mentioned in this step (i.e. Step: 2) for collating harmful prompts from the SFT model you fine-tuned previously.

Query: How can you obtain $\theta_{\text{base}(\sim)}$?

Response:

- (i) Prompt your SFT model (individually for PEFT and Full-FT) received from **Step: 1** on the dataset mentioned in **Step: 2**. Use zero-shot prompting up here. Use the “***prompt***” column in the dataset for this purpose.
- (ii) Then take LLM as judge: Use [meta-llama/Llama-2-7b-chat-hf](#) checkpoint as LLM judge for your assignment. Run it in *16-bit precision (bf16 is the best option)* and *don't quantize* this model. Prompt this LLM judge on the output received when you prompted SFT model in zero-shot setting (refer **(i)**). Here, prompts for LLM as judge must be formed to output answers telling whether the output received by the SFT model is *harmful* or *not-harmful*. It should be a zero-shot prompting.
- (iii) Remove ambiguous cases and select only 100 samples that are returned as *harmful* by your LLM as judge.

As a next step, use the obtained 100 samples which your model has predicted to be *harmful* as samples to generate δ_{safe} . To this purpose, fine-tune $\theta_{\text{base}(+)}$ on these 100 *harmful* detected samples to obtain $\theta_{\text{base}(\sim)}$. For this fine-tuning, use the “***prompt***” column as your harmful input and “***chosen***” column as your safe output.

Finally, obtain δ_{safe} as per the equation given above.

(Step: 3) Add safety vector

Goal: $\delta_{\text{safe}} + \text{SFT}$ and $\delta_{\text{safe}} + \text{SFT} + \text{DARE}$

Models from this step: SFT + RESTA and SFT + DARE + RESTA

Write your own code to perform model vector addition to obtain SFT + RESTA & SFT + DARE + RESTA from SFT and SFT + DARE respectively. You can use the [mergekit](#) library for this purpose.

(Step: 4) Generate and score responses

Dataset for [unsafety](#) evaluation: [SoftMINER-Group/HarmEval](#)

Prompt all the model variants mentioned in **Objective_1** and obtained iteratively through **Steps: 1 to 3** on this dataset.

Evaluation-

(i) Unsafety (Objective_1):

LLM as judge: Use [meta-llama/Llama-2-7b-chat-hf](#) checkpoint as LLM judge for your assignment. Run it in *16-bit precision (bf16 is the best option)* and *don't quantize* this model. For the outputs generated for each setup on the above mentioned dataset (**in Step: 4**), use this LLM judge to detect whether the output is *harmful* or *no-harmful*. Use the same prompt you used in **Step: 2**. Then calculate and report the [unsafety](#) score for each model setups. You have to do this on all five setups, one model and two fine-tuning strategies.

(ii) Performance (Objective_2):

Model comparison for performance evaluation will be performed on the last **10%** samples (**as already covered in Step: 1**). You have to do this on all five setups, one model and two fine-tuning strategies. Use ROUGE-L, METEOR and BLEU metrics for this evaluation.

Helpful resources

[question] task_arithmetic simple question

<https://github.com/arcee-ai/mergekit/issues/438>

DARE: Separate base models for delta measurement and merge?

<https://github.com/arcee-ai/mergekit/issues/25>

any example yml file for model_stock merging method?

<https://github.com/arcee-ai/mergekit/issues/270>

Can I perform a task_arithmetic or TIES merge on two embedding matrices?

<https://github.com/arcee-ai/mergekit/issues/27>

Could you please explain how passthrough slicing works?

<https://github.com/arcee-ai/mergekit/issues/53>

Request for scripts to reproduce the results in the Mergekits paper

<https://github.com/arcee-ai/mergekit/issues/261>

Can you please explain the weight and density parameters related to the TIES merging?

<https://github.com/arcee-ai/mergekit/issues/5>

Something which is a bit advanced

Question about the implement of sign election in TIES

<https://github.com/arcee-ai/mergekit/issues/474>

What is the difference between mergekit-moe and mergekit-yaml

<https://github.com/arcee-ai/mergekit/issues/151>